Project Status and Next Steps

Formula 1 Podium Prediction · CSEN 903, Milestone 1

> **As of:** 13 October 2026
> **Deadline:** 18 October 2026, 11:59 PM (5 days left)
> **To submit:** Kaggle notebook link, GitHub repository link, report PDF

1. Where we stand

Nine of the ten plan steps are finished, and most of the tenth. The notebook holds the whole pipeline (Tasks 1–38 plus a closing section), and a 22-page report PDF exists as a draft. The project is ahead of the plan's timeline, which had integration and the report on 15–16 October.

What is left cannot be done from the code alone. It needs the team: running the notebook on Kaggle, completing and reviewing the report, approving the open decisions, rehearsing, and submitting.

Committed and pushed	Steps 1 to 9 (Tasks 1–38)
Committed and pushed	Step 10 work: notebook section 11 with a run check, report draft, README, `requirements.txt`, `.gitignore`, path fixes
Not started	Kaggle Run All, team review of the report, decision approvals, rehearsal, submission

2. Finished

Step	Owner	Tasks	What was produced	Key result
Shared components	All		Config, plot style, leakage guard, data contract, decision log	One place for the seed, split years and blacklist
1. Data audit	M1	1–3	Table inventory, key checks, six "before" figures	16 keys unique; 22 foreign keys with no orphans
2. Data cleaning	M1	4–8	`clean_df`, five "after" figures	26,759 raw rows → 24,575, one per driver per race
3. Data-engineering questions	M2	9–11	Answers, figures and justifications for Q1–Q3	No home advantage once grid is held fixed (odds ratio 0.95)
4. Feature engineering	M3	12–14	`model_df` with 21 features in 6 groups, feature registry, two leakage tests	Both leakage tests pass
5. Split and preprocessing	M3	15–20	Split by season, pipeline, training-window and two order experiments	Order of preprocessing steps makes no measurable difference
6. Modeling	M4	21–28	Grid rule, logistic regression, random forest, LightGBM, neural network	Test PR-AUC 0.701–0.711 for learned models; 0.647 for the grid rule
7. Ablation	M3	29–31	Per-group and per-block retraining, two figures	Starting position –0.12 on test when removed; context 0.00
8. Explainable AI	M2	32–35	Permutation importance, SHAP, two case studies, LIME, six figures	`grid` carries 41% of SHAP magnitude
9. Inference function	M4	36–38	`predict_podium()`, score calibration, two figures	Reproduces stored features and scores for 36 held-out rows

Step 10 items already done

Item	Where
Limitations and future work, with a requirement-to-section table	Notebook, section 11
Run Check: headline numbers as quoted against the current run	Last cell of the notebook
Report draft, 22 pages	`report/report.pdf` (source `report/report.md`, rebuilt with `python report/build_report.py`)
Repo setup	`README.md`, `requirements.txt`, `.gitignore`
Machine-specific paths removed	Config cell of the notebook; two docs
Docs brought up to date	`docs/notebook_guide.md`, `docs/status_and_next_steps.md`, `docs/decision_log.md` (DEC-001 to DEC-033)

Deliverables of the brief

Deliverable	Status
1. Run-all notebook	Runs clean locally. **Not yet run on Kaggle.**
2. EDA before and after cleaning	Done (11 figures)
3. Modeling table, one row per `(raceId, driverId)`	Done
4. Model comparison table on all three splits	Done
5. Ablation and two preprocessing-order experiments	Done
6. XAI suite, global and local	Done
7. Inference function on complete and incomplete records	Done
8. Report PDF	Draft. Team names and Kaggle link missing.

3. Next steps

In the order they should happen.

#	What	Who	By	Done when
1	Pull `main`	Everyone	Tue 13	Each member has `report/`, `README.md` and notebook section 11 locally
2	First Run All on Kaggle	M1	Tue 13	The notebook finishes with no error and its last cell, Run Check, says all headline numbers match
3	Fill in the team names and IDs and the Kaggle link in `report/report.md`; rebuild the PDF	M1	Wed 14	No placeholder text on page 1 of the PDF
4	Each owner reviews the notebook step and report section written for them	M2, M3, M4	Wed 14	Owner can explain every number and decision in it
5	Approve or change the 19 decisions still marked "Proposed"	All	Thu 15	Every row in the decision log reads "Approved" or "Rejected"
6	Code freeze and final Kaggle Run All	M1	Fri 16	Final outputs saved; Kaggle link set to accessible
7	Final report PDF	All	Sat 17	PDF rebuilt after the last change and read once end to end

#	What	Who	By	Done when
8	Rehearsal: 10 minutes per member, each questioned on every stage	All	Sat 17	Every member has answered "why" questions on a stage that is not theirs
9	Submit through the form	One named person	Sun 18	Kaggle link, GitHub link and PDF sent before 11:59 PM

Details for the steps that need care

Step 2, Kaggle Run All

- Add the dataset "Formula 1 World Championship (1950–2024)" to the notebook. The config cell looks under `/kaggle/input` by itself.
- Turn Internet on in the notebook settings. The notebook installs `shap` and `lime` itself if the Kaggle image does not have them.
- **Read the last cell, Run Check.** The result paragraphs in the notebook and the whole report quote numbers from the local run. A newer dataset version or different library versions can shift them. Run Check lists 25 headline numbers as quoted next to the values of the run and says which differ.
- **If counts differ** (rows, features), the Kaggle dataset is a different version. **If only scores differ**, a library behaves differently, most likely for the neural network. In both cases the text that quotes the number has to be corrected.

Step 4, who reviews what

Member	Notebook	Report sections	Decisions
M1	Steps 1–2, section 11	1, 2, 10	DEC-009, DEC-010
M2	Steps 3 and 8	3, 8	DEC-012, DEC-032
M3	Steps 4, 5 and 7	4, 5, 7	DEC-015 to DEC-021, DEC-031
M4	Steps 6 and 9	6, 9	DEC-022, 023, 025–027, 029, DEC-033

Step 5, the two decisions to settle first

- **DEC-009 (non-starters)** and **DEC-010 (pit-lane starts)** decide which rows are in the modeling table. Changing either changes the row counts and every number after them, so they must be settled before the final Kaggle run.
- **DEC-029 (LightGBM as the selected model)** is the one most likely to be asked about. On test it equals the logistic regression (0.709 against 0.711). The notebook and report say so openly.

4. Risks

- **Kaggle is untested.** This is the largest remaining risk, because the submission goes through it. Do the first run today, not on Friday.
- **Numbers are typed into the text, not computed.** If the Run Check on Kaggle reports differences, the notebook paragraphs and the report must be corrected by hand.
- **Steps 7, 8 and 9 were written in one sitting, not by their owners.** Every member has to defend every stage, so the reviews in step 4 are not optional.
- **One sentence in the report rests on outside knowledge:** that the 2020 Italian Grand Prix was decided by a safety car, a red flag and a penalty. Confirm it or cut it.
- **The test seasons have been used.** Any change to the model or threshold from here on that is judged on test scores would weaken the evaluation. Do not retune.

5. Submission checklist

- Kaggle notebook: runs top to bottom, link accessible to the graders
- GitHub repository: `main` pushed, link accessible
- Report PDF: team names and Kaggle link filled in, rebuilt after the last change
- Form submitted before 18 October, 11:59 PM